Adaptive Task Allocation for Heterogeneous Bio-Inspired Agents

Final Year B.Tech Project Synopsis
Department of Computer Science and Engineering
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1 Problem Statement

Large language model based multi-agent systems divide complex work among interacting agents. A single task may require exploration, sequential tool use, independent execution, comparison of alternatives, and recovery from failure. These operations impose different requirements on communication, memory, and control. A fixed workflow can therefore assign an unsuitable mechanism to a subtask, duplicate reasoning, or repeatedly involve a coordinator when local execution would suffice. The resulting overhead can increase token consumption and completion time without improving the correctness of the final result.

Existing research provides a basis for adaptive coordination. MasRouter selects collaboration modes, roles, and language models according to the task [1]. AMRO-S applies ant-colony mechanisms to routing and maintains task-specific pheromone memories [2]. Earlier optimization research also combines different animal-inspired algorithms through performance-dependent switching [3]. These studies motivate adaptive selection and establish the foundations on which this project builds. The research question is whether selecting among explicitly different biological coordination mechanisms provides measurable value beyond a conventional specialist-agent system.

Biological research supplies candidate mechanisms for this investigation. Honeybee swarms combine evidence and inhibit competing alternatives during collective decisions [4]. Octopus arm control demonstrates a division between central direction and peripheral execution [5]. Crow metatool use illustrates sequences in which an intermediate tool enables a later action [6]. A brittle-star-inspired robot demonstrates adaptation after physical damage [7]. Together, these findings motivate distinct approaches to decision-making, local autonomy, tool sequencing, and recovery. Their translation into software provides a concrete basis for designing agents with different operating mechanisms.

This project proposes a human-inspired AI coordinator that decomposes an incoming task, records dependencies, and dynamically allocates ready subtasks to a pool of bio-inspired agents. The coordinator implements planning, delegation, monitoring, and reallocation. The proposed pool includes ant, bee, octopus, crow, wolf, elephant, and starfish-inspired modules. Their intended functions are reinforced search, evidence aggregation, local execution, tool sequencing, cooperative exploration, episodic recall, and failure recovery respectively. A common interface will allow the modules to exchange results and operate within a shared computational budget. The animal names describe design inspirations; their suitability will be determined by computational behavior.

The central challenge is selecting an appropriate mechanism for each subtask while accounting for routing errors, communication overhead, and limited computation. The coordinator will use observable task features and recorded outcomes to estimate suitability. Development will begin with ant, bee, and octopus mechanisms before extending the framework to additional modules. The system will be evaluated on verifiable planning, tool-use, and reasoning tasks that expose differences in exploration, coordination, and execution requirements.

The project will investigate whether adaptive allocation improves task success, efficiency, or recovery compared with fixed workflows and conventional specialist agents. Comparisons will include a single agent, the strongest individual mechanism, fixed allocation, random allocation, and a conventional coordinator with equivalent resources. Held-out tasks, repeated runs, and component ablations will help distinguish the effects of mechanism diversity and routing from those of additional computation. The expected research contribution

is a modular coordination framework and an empirical account of the conditions under which its mechanisms are useful.

2 Proposed Approach

The framework will use a shared language-model backbone where practical. Differences between agents will be implemented through control flow, communication rules, memory updates, and stopping conditions. The coordinator will maintain task dependencies, execution status, available budget, and performance records.

Agent	Proposed mechanism
Ant	Search over candidate paths with outcome-based reinforcement and evaporation [2].
Bee	Independent evidence accumulation, inhibition of weaker alternatives, and threshold-based selection [4].
Octopus	Bounded autonomous execution using local worker state and summary reporting [5].
Crow	Planning of tool prerequisites and intermediate actions [6].
Wolf	Cooperative exploration with diversity constraints to reduce duplicate search; an exploratory extension.
Elephant	Retrieval of previous task episodes and outcomes to support subsequent decisions; an exploratory extension.
Starfish	Checkpointing and reassignment following worker failure, drawing on brittle-star-inspired robotics [7].

The workflow comprises task decomposition, mechanism selection, subtask execution, result verification, and feedback-based reallocation. The starfish module uses an analogy to brittle-star adaptation; brittle stars and starfish are distinct biological groups.

3 Research Questions and Possible Solutions

RQ1: Do the mechanisms have complementary strengths?

Possible solution: Evaluate each implemented mechanism on the same task families. Construct capability profiles using correctness, cost, and latency, then identify the task characteristics associated with each mechanism's strengths.

RQ2: Can dynamic allocation improve performance within a fixed budget?

Possible solution: Begin with routing rules based on task features, then investigate a lightweight selector learned from recorded outcomes. Compare dynamic allocation against fixed routing, random routing, and the strongest individual mechanism.

RQ3: What value do the biological mechanisms add?

Possible solution: Compare the framework with conventional specialist agents using equivalent models, tools, memory access, and budgets. Remove reinforcement, inhibition, and local autonomy separately to measure their individual contributions.

RQ4: Can the framework adapt to failures and changing task requirements?

Possible solution: Introduce controlled worker timeouts and tool failures. Evaluate checkpoint-based reassignment and update routing preferences from execution feedback. Test generalization on held-out task

distributions.

RQ5: When does coordination become inefficient?

Possible solution: Measure coordinator tokens, communication volume, and elapsed time separately. Introduce budget limits, stopping conditions, and a direct single-agent route for simple tasks.

4 Expected Outcomes

- **A modular research prototype:** An AI coordinator with a common interface for heterogeneous bio-inspired agents, initially validated with three mechanisms and designed to accommodate the seven proposed modules.
- **Task-specific capability profiles:** Evidence identifying which mechanisms are suitable for different task classes and when their combination is useful.
- **Comparative performance results:** Measurements of task success, token consumption, inference cost, completion time, and recovery after controlled failures.
- **Component-level analysis:** Ablation studies separating the effects of adaptive routing, memory, communication, and agent-specific mechanisms.
- **Reproducible research materials:** Documented code, configurations, evaluation scripts, and experiment logs supporting the final report and a potential conference submission.

The working hypothesis is that adaptive allocation can improve task success at an equal computational budget, or reduce cost while maintaining comparable success. The study will report the conditions supporting this hypothesis, together with limitations and cases where simpler methods perform better.

5 Mapping to Sustainable Development Goals

In accordance with DTU Order No. 106/DTU/Dean Acad (UG)/2025-26/265 dated 21/08/2026, the project is mapped to the following Sustainable Development Goals and specific targets. The contribution and impact mapping is subject to verification by the project supervisor.

SDG	Target	Contribution and impact of the project
SDG 9 Industry, Innovation and Infrastructure *Primary*	9.5	Target 9.5 concerns scientific research and technological capability. The project contributes through an implemented AI coordination framework, reproducible experiments, and documented findings. It develops technical knowledge about adaptive agent orchestration and provides a foundation for further research and potential applications.
SDG 8 Decent Work and Economic Growth *Secondary*	8.2	Target 8.2 concerns productivity through technological upgrading and innovation. The project examines whether planning and tool-use tasks can be completed correctly with less time or computational expenditure. Its potential productivity contribution depends on measured improvements and subsequent validation in practical workflows.

Assessment of contribution: The project will track successful tasks per unit cost, completion time, and reproducibility of its research outputs. These are project-level measures rather than official national SDG indicators. Energy or carbon reductions will not be inferred from token savings alone.

Official target sources: United Nations, SDG 9 and SDG 8.

References and Research Basis

[1] Y. Yue, G. Zhang, B. Liu, G. Wan, K. Wang, D. Cheng, and Y. Qi, "MasRouter: Learning to Route LLMs for Multi-Agent Systems," in *Proceedings of ACL*, 2025, pp. 15549–15572. [Paper]

Basis: Supports task-dependent selection of collaboration modes, agent roles, and language models. Establishes a computational foundation for the proposed coordinator.

[2] X. Wang et al., "Efficient and Interpretable Multi-Agent LLM Routing via Ant Colony Optimization," *arXiv preprint arXiv:2603.12933*, 2026. [Paper]

Basis: Provides a direct precedent for pheromone-based routing, task-specific routing memory, and outcome-based adaptation in heterogeneous LLM systems.

[3] R. Damaševičius and R. Maskeliūnas, "Agent State Flipping Based Hybridization of Heuristic Optimization Algorithms: A Case of Bat Algorithm and Krill Herd Hybrid Algorithm," *Algorithms*, vol. 14, no. 12, art. 358, 2021. [Paper]

Basis: Supports hybridization of different animal-inspired algorithms and performance-dependent switching. Establishes relevant prior work for combining biological mechanisms.

[4] T. D. Seeley, P. K. Visscher, T. Schlegel, P. M. Hogan, N. R. Franks, and J. A. R. Marshall, "Stop Signals Provide Cross Inhibition in Collective Decision-Making by Honeybee Swarms," *Science*, vol. 335, no. 6064, pp. 108–111, 2012. [Paper]

Basis: Provides biological and modeling support for evidence accumulation, inhibition of competing alternatives, and resolution of decision deadlock.

[5] G. Sumbre, Y. Gutfreund, G. Fiorito, T. Flash, and B. Hochner, "Control of Octopus Arm Extension by a Peripheral Motor Program," *Science*, vol. 293, no. 5536, pp. 1845–1848, 2001. [Paper]

Basis: Motivates local execution under higher-level direction. Its motor-control findings provide biological inspiration for bounded worker autonomy.

[6] A. H. Taylor, G. R. Hunt, J. C. Holzhaider, and R. D. Gray, "Spontaneous Metatool Use by New Caledonian Crows," *Current Biology*, vol. 17, no. 17, pp. 1504–1507, 2007. [Paper]

Basis: Motivates explicit intermediate goals and tool dependencies, where one action enables the tool required for a later action.

[7] T. Kano et al., "A Brittle Star-Like Robot Capable of Immediately Adapting to Unexpected Physical Damage," *Royal Society Open Science*, vol. 4, art. 171200, 2017. [Paper]

Basis: Provides an implemented robotics precedent for adaptation after damage. It motivates investigation of recovery and reallocation mechanisms in the proposed software framework.

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